

Derivation of partial differential eqⁿ by the elimination of arbitrary constants

If z is a dependent variable and x, y are independent.

$$p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y}, \quad r = \frac{\partial^2 z}{\partial x^2}, \quad s = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}, \quad t = \frac{\partial^2 z}{\partial y^2}$$

Q) form the partial diffⁿ equation by eliminating constants from $z = ax + by + a^2 + b^2$.

Sol: diff w.r.t x | diff w.r.t y then P.D.E is

$$p = \frac{\partial z}{\partial x} = a \quad \left| \quad \frac{\partial z}{\partial y} = b = q \right. \quad \boxed{z = px + qy + p^2 + q^2}$$

Q) Eliminate a & b from $z = axe^{ay} + \frac{1}{2}a^2e^{2y} + b$.

$$\frac{\partial z}{\partial x} = ae^y, \quad \frac{\partial z}{\partial y} = axe^y + a^2e^{2y} = ax\left(\frac{\partial z}{\partial x}\right) + \left(\frac{\partial z}{\partial x}\right)^2$$

$$\boxed{q = ap + p^2}$$

Lagrange's partial diffⁿ eqⁿ: A diffⁿ equation of the form $PP + qQ = R$ where

P, Q, R are functions of x, y, z , solⁿ of this diffⁿ eqⁿ is $f(u, v) = 0$
(or) $f(a, b) = 0$

where

$$P = \left(\frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial y} \right), \quad Q = \left(\frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial z} \right)$$

$$R = \left(\frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x} \right)$$

Working Rule:

For the solⁿ of partial diffⁿ equation $PP + qQ = R$, form A.E

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

find 2 independent integrals of auxiliary equation say $u(x, y, z) = a, v(x, y, z) = b$
then the general integral of the eqⁿ is given by $f(u, v) = 0$ (or) $u = \phi(v)$ (or) $v = \psi(u)$

$$f(a, b) = 0 \quad \text{(or)} \quad a = g(b) \\ b = h(a)$$

Q) Solve $P+Q=z/a \Rightarrow P+Q=R$
 $P=1, Q=1, R=z/a.$

A.E $\Rightarrow \frac{dx}{1} = \frac{dy}{1} = \frac{dz}{z/a}$
 $\Rightarrow \frac{dx=dy}{x-y=c} \quad \int dy = \int \frac{a}{z} dx \Rightarrow y = a \ln z + c' \Rightarrow y - a \log z = c'$
 $y/a = \log(z/d) \Rightarrow z = \frac{1}{d} e^{y/a}$

$k = z e^{-y/a} \quad (c'') \quad z = k e^{y/a}.$

$f(c, k) = 0$

$f(x-y, z e^{-y/a}) = 0$

Q) Solve $(mz-ny)P + (mx-ly)Q = ly-mx$

Sol: $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{ldx+m dy+ndz}{lP+mQ+nR}.$

$\frac{dx}{mz-ny} = \frac{dy}{mx-ly} = \frac{dz}{ly-mx} = \frac{ndx+ydy+zdz}{0} = \frac{ldx+m dy+ndz}{0}$

$ndx+ydy+zdz=0 \quad | \quad (ii) \quad lx+my+nz=b$
 $x^2+y^2+z^2=a.$

$f(a,b)=0.$

$f(x^2+y^2+z^2, lx+my+nz)=0.$

Q) Eliminate arbitrary function of form $z=f(x^2-y^2)$

Sol: $z=f(x^2-y^2)$

$\frac{\partial z}{\partial x} = f'(x^2-y^2)(2x)$

$f(a,b)=0 \rightarrow b=f(a)$

$\frac{\partial z}{\partial y} = f'(x^2-y^2)(-2y)$

$f(u,v)=0 \Rightarrow v=f(u)$

$\frac{P}{Q} = -x/y \Rightarrow Py+Qx=0$

$\phi(z, x^2-y^2).$

$P = \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} = (0 \cdot 0 - 1(-2y)) = 2y$

$$Q = \frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial z} = 2x; \quad R = \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x} = 0$$

$$P^2 + Q^2 = R$$

$$P(2y) + Q(2x) = 0 \Rightarrow \boxed{Py + Qx = 0}$$

a) Form a PDE by eliminating arbitrary constant $\phi(x^2 + y^2 + z^2, z - 2xy) = 0$

solⁿ: find P, Q, R on solving $(P - Q)z = yx$.

Non-linear partial differential equations.

$$f(x, y, z, P, Q) = 0 \rightarrow (i)$$

Let there be any other condition $F(x, y, z, P, Q) = 0 \rightarrow (ii)$

Then solving P, Q from (i) & (ii) and substitute in $dz = Pdx + Qdy$

standard forms:

1) If the PDE of the form $f(P, Q) = 0$ then

$$z = ax + by + c$$

$$z = ax + \phi(a)y + c$$

a) Solve $P^2 + Q^2 = 1$, this is in the form $f(P, Q) = 0$

Replacing P, Q by a, b

$$a^2 + b^2 = 1 \Rightarrow b = \sqrt{1 - a^2}$$

$$\underline{z = ax + by + c = ax + \sqrt{1 - a^2}y + c}$$

2) If the eqⁿ of form $f(z, P, Q) = 0$, let us assume $z = \phi(x + ay)$

put $x + ay = X$

then $z = \phi(X)$

$$P = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial X} \cdot \frac{\partial X}{\partial x} = \frac{\partial z}{\partial X} \cdot 1 = \frac{\partial z}{\partial X} \quad \left| \quad Q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial X} \cdot \frac{\partial X}{\partial y} = a \frac{\partial z}{\partial X}$$

Then (i) can be written as $f(z, \frac{\partial z}{\partial X}, a \frac{\partial z}{\partial X}) = 0$.

which is ordinary diffⁿ eqⁿ of 1st order where $X = x + ay$

3) solve $P^2 = zQ$

Putting $z = \phi(x + ay)$ $P = \frac{\partial z}{\partial x}$, $Q = a \frac{\partial z}{\partial x}$

$$\left(\frac{\partial z}{\partial x}\right)^2 = z \left(a \frac{\partial z}{\partial x}\right) \Rightarrow \frac{d}{dx} z = az$$

$$\frac{dz}{z} = a dx$$

$$\Rightarrow \log z = ax + \log b \Rightarrow z = be^{ax} = be^{a(x+ay)}$$

Q) Solve $q(p^2z + q^2) = 4$

$f(z, p, q) = 0$, Now putting $z = \phi(x+ay) = \phi(x)$ $x = x+ay$

Then $p = \frac{dz}{dx}$, $q = a \frac{dz}{dx}$

$$q \left(\left(\frac{dz}{dx} \right)^2 z + a^2 \left(\frac{dz}{dx} \right)^2 \right) = 4 \Rightarrow q \left(\frac{dz}{dx} \right)^2 (z + a^2) = 4$$

$$\frac{3 \frac{dz}{dx}}{\frac{dz}{dx}} \sqrt{z+a^2} = 2$$

$$\frac{3(z+a^2)^{3/2}}{3/2} = 2 \int dx + b \Rightarrow \boxed{(z+a^2)^{3/2} = x+b}$$

③ If $f(x, p) = F(y, q)$; $p = f_1(x, a)$ $q = f_2(y, a)$

then $dz = p dx + q dy$

$$z = \int f_1(x, a) dx + \int f_2(y, a) dy + c$$

Q) Solve $p^2 + q^2 = x+y$

Sol: $p^2 - x = y - q^2 = a \Rightarrow p = \sqrt{x+a}$, $q = \sqrt{y-a}$ $\{ dz = p dx + q dy$

$$dz = \int \sqrt{x+a} dx + \int \sqrt{y-a} dy = \frac{2}{3} (x+a)^{3/2} + \frac{2}{3} (y-a)^{3/2} + c$$

Q) Solve $pq = xy$

Sol: $\frac{p}{x} = \frac{y}{q} = a$ $p = ax$, $q = y/a$

$$dz = p dx + q dy = \frac{1}{3} a x^3 + \frac{1}{3} \frac{y^3}{a} \Rightarrow \boxed{z = \frac{ax^3}{3} + \frac{y^3}{3a} + c}$$

④ $z = px + qy + f(p, q)$

Sol: $z = ax + by + f(a, b)$

Q) $z = px + qy + c \sqrt{1+p^2+q^2}$

$$\boxed{z = ax + by + c \sqrt{1+a^2+b^2}}$$

Charpit's Method:

If $f(x, y, z, p, q) = 0$ and it written in another form $F(x, y, z, p, q) = 0$ then

$$\text{A.E.} \quad \frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{df}{0}$$

find p & q and substitute in $dz = p dx + q dy$.

Q) Solve. $(p^2 + q^2)y = qz \rightarrow (i)$

A.E. $f = (p^2 y + q^2 y - qz) = 0$

$$\text{A.E.} \quad \frac{dp}{-pq} = \frac{dq}{(p^2 y + q^2 y) + q(-z)} = \frac{dz}{-p(2py) - q(2qy - z)} = \frac{-dx}{2py} = \frac{-dy}{2qy - z}$$
$$\downarrow \quad \downarrow$$
$$\frac{dq}{p^2} \quad \frac{-dz}{2y(p^2 + q^2) - qz} \Rightarrow \frac{-dz}{2qz - qz} = \frac{-dz}{qz}$$

taking it as parts.

$$p dp = -q dq \Rightarrow p^2 + q^2 = a^2 \rightarrow (ii)$$

Now from (i) and (ii)

$$a^2 y = qz \Rightarrow q = \frac{a^2 y}{z}$$

Now substitute q in eqⁿ (ii) we get

$$p = \sqrt{a^2 - q^2}$$
$$p = a \sqrt{1 - \frac{a^2 y^2}{z^2}}$$

$$dz = p dx + q dy \Rightarrow dz = \left(a \sqrt{1 - \frac{a^2 y^2}{z^2}} \right) dx + \frac{a^2 y}{z} dy$$

$$z dz = a \left(\sqrt{z^2 - a^2 y^2} \right) dx + a^2 y dy$$

$$(1) \quad \frac{z dz - a^2 y dy}{\sqrt{z^2 - a^2 y^2}} = a dx$$

Integrating we have $\sqrt{z^2 - a^2 y^2} = ax + b$

$$\boxed{z^2 = a^2 y + (ax + b)^2} \text{ Ans}$$

Q1 Solve $ax^2 - px^2 - 2qxy + pq = 0$.

Ans: $z = ay + b(x^2 - a)$.

Standard forms.

(i) If $f(p, q) = 0$.

$$\frac{dp}{0} = \frac{dq}{0} = \frac{dz}{---} = \frac{dx}{---} = \frac{dy}{---} \Rightarrow \begin{aligned} dp=0 &\Rightarrow p=a \\ dq=0 &\Rightarrow q=b. \end{aligned}$$

$$dz = p dx + q dy$$

$$dz = a dx + b dy \Rightarrow$$

$$\boxed{\begin{aligned} z &= ax + by + C \\ z &= ax + \phi ay + C \end{aligned}}$$

(ii) If $f(z, p, q) = 0$

$$\frac{dp}{p \frac{\partial f}{\partial z}} = \frac{dq}{q \frac{\partial f}{\partial z}} = \frac{dz}{---} = \frac{dx}{---} = \frac{dy}{---} \Rightarrow \frac{dp}{p} = \frac{dq}{q}$$

$$\log p = \log q + \log a$$

$$p = aq$$

Now $dz = p dx + q dy$

$$dz = p dx + a p dy = p(dx + a dy) = p d(x + ay)$$

$$= p dx \quad (\text{where } x = x + ay)$$

$$dz = p dx$$

$$p = \frac{dz}{dx} \quad \& \quad q = a p = a \frac{dz}{dx} \quad \text{then } f\left(z, \frac{dz}{dx}, a \frac{dz}{dx}\right) = 0.$$

(iii) If $f(x, p) = F(y, q) = 0$

$$\frac{dp}{\frac{\partial f}{\partial x} = 0} = \frac{dq}{\frac{\partial F}{\partial y} = 0} = \frac{dz}{-\frac{\partial f}{\partial p} - \frac{\partial F}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{\frac{\partial F}{\partial q}}$$

Now taking (i) and (ii) part $\frac{\partial f}{\partial p} dp + \frac{\partial f}{\partial x} dx = 0$.

$$\Rightarrow df = 0$$

$$\Rightarrow f = a$$

$$f(x, p) = F(y, q) = a$$

$$(iv) \quad z = px + qy + f(p, q) = 0$$

$$f = px + qy + f(p, q) - z = 0$$

$$\frac{dp}{p \cdot p} = \frac{dq}{q \cdot q} = \frac{dz}{-z} = \frac{dx}{-x} = \frac{dy}{-y}$$

$$\text{from 1st 2} \quad dp = 0, dq = 0$$

$$p = a, q = b$$

$$\boxed{z = ax + by + f(a, b)}$$

PDE of second orders i.e. $f(x, y, z, p, q, r, s, t) = 0$.

Most general form

$$Rr + Ss + Tt + Pp + Qq + Zz = F \quad \left(\text{where } R, S, T, P, Q, Z \text{ are functions of } x, y \right)$$

p, q, r, s, t

$$\text{Solve: } S = 2x + ay$$

$$\frac{\partial^2 z}{\partial x \partial y} = 2x + ay$$

Integrating w.r.t x then

$$\frac{\partial z}{\partial y} = x^2 + ayx + f(y)$$

where f is an arbitrary function.

Integrating w.r.t y then

$$z = x^2y + \frac{a}{2}y^2x + \int f(y)dy + g(x)$$

$$\boxed{z = x^2y + y^2x + F(y) + g(x)}$$

$$\text{Solve } P + r + s = 1$$

$$\frac{\partial z}{\partial x} + \frac{\partial^2 z}{\partial x^2} + \frac{\partial q}{\partial x} = 1$$

$$\text{Integrating w.r.t } x \quad \therefore z + p + q = x + f(y)$$

$$p + q = x + f(y) - z$$

$$\text{AE } \frac{dx}{1} = \frac{dy}{1} = \frac{-dz}{x + f(y) - z}$$

$$\text{1st two } x = y + a \quad \text{i.e. } \boxed{x - y = a}$$

taking last two

$$\frac{dy}{1} = \frac{dz}{x + f(y) - z}$$

$$\frac{dz}{dy} = x + f(y)z$$

$$\frac{dz}{dy} + z = x + f(y) \Rightarrow \text{which is linear} \quad I.F = e^{\int dy} = e^y$$

$$ze^y = \int e^y (x + f(y)) dy + b$$

$$= \int e^y (a + y + f(y)) dy + b = \int a e^y dy + \int y e^y dy + \int f(y) e^y dy + b$$

$$= a e^y + \phi(y) + b$$

$$(or) \quad z = a + e^{-y} \phi(y) + b e^{-y}$$

$$\boxed{z = (x-y) + e^{-y} \phi(y) + F(x-y) e^{-y}}$$

Classification of 2nd order partial differential equations:

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + f(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}) = 0 \rightarrow (*)$$

Let the 2nd order diffⁿ equation is this (*), the diff eqns (*)

a) Elliptic, if $B^2 - 4AC < 0$

b) Hyperbolic, if $B^2 - 4AC > 0$

c) Parabolic, if $B^2 - 4AC = 0$

Here A, B, C are constants or functions of x & y.

Q) Classify the following differential equation a) $\frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial x^2}$

b) $\frac{\partial^2 u}{\partial t^2} - 4 \frac{\partial^2 u}{\partial t \partial x} + \frac{\partial^2 u}{\partial x^2}$

c) $\frac{\partial^2 u}{\partial t^2} + 4 \frac{\partial^2 u}{\partial x \partial t} + 4 \frac{\partial^2 u}{\partial x^2}$

Sol! a) $\frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial x^2}$

Here $A = B = C = 1$

$$B^2 - 4AC = -3$$

$B^2 - 4AC < 0$ this is elliptic

b) $A = 1, B = -4, C = 1$ then

$$B^2 - 4AC$$

$$16 - 4 = 12$$

$B^2 - 4AC > 0$ (Hyperbola)

c) $A = 1, B = 4, C = 4$

$$B^2 - 4AC$$

$$16 - 16 = 0$$

parabolic.

Q) find whether the following are hyperbolic, parabolic & elliptic

i) $\frac{\partial^2 u}{\partial t^2} + t \frac{\partial u}{\partial x \partial t} + x \frac{\partial^2 u}{\partial x^2}$ $A=1, B=t, C=x$
 $B^2 - 4AC = t^2 - 4x$

If $t^2 - 4x < 0$ (or) $t^2 < 4x$ then given operator is elliptic
 If $t^2 - 4x > 0$ (or) $t^2 > 4x$ the given operator is hyperbolic
 $t^2 - 4x = 0$ (or) $t^2 = 4x$ parabolic

(ii) $x \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} + u$ $A=x^2, B=0, C=-1$
 $B^2 - 4AC = 0 - 4x^2(-1) = 4x^2$
 x^2 always positive: $B^2 - 4AC = 4x^2 > 0$ hyperbolic
 If $x=0$ Parabolic.

(iii) $t \frac{\partial^2 u}{\partial t^2} + 2 \frac{\partial^2 u}{\partial x \partial t} + x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x}$ $\rightarrow A=t, B=2, C=x$
 $B^2 - 4AC = 4 - 4tx$
 $B^2 - 4AC = 4(1 - xt)$
 If $1 - xt > 0$ (or) $1 > xt$ Hyperbolic
 $1 - xt < 0$ $1 < xt$ elliptic
 $1 - xt = 0$ $xt = 1$ parabolic.

PDE with const coefficient:

A diff eqⁿ of the form

$$\left(A_0 \frac{\partial^n z}{\partial x^n} + A_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + A_2 \frac{\partial^n z}{\partial x^{n-2} \partial y^2} + \dots + A_n \frac{\partial^n z}{\partial y^n} \right) + \left(B_0 \frac{\partial^{n-1} z}{\partial x^{n-1}} + B_1 \frac{\partial^{n-1} z}{\partial x^{n-2} \partial y} + \dots \right) + \dots + \left(M \frac{\partial z}{\partial x} + N \frac{\partial z}{\partial y} \right) + Bz = f(x, y)$$

where $A_0, A_1, \dots, B_0, B_1, \dots, M, N, B$ are constants

solution of linear homogenous partial differential equations.

$$A_0 \frac{\partial^n z}{\partial x^n} + A_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + \dots + A_n \frac{\partial^n z}{\partial y^n} = f(x, y)$$

Denoting $\frac{\partial}{\partial x}$ by D , $\frac{\partial}{\partial y}$ by D' then.

$$(A_0 D^n + A_1 D^{n-1} D' + A_2 D^{n-2} D'^2 + \dots + A_n D'^n) z = f(x, y)$$

$$\phi(D, D') z = f(x, y)$$

$$\text{If } \phi(D, D') = (D - m_1 D')(D - m_2 D') \dots (D - m_n D')$$

$$\text{The } \phi(D, D') = A_0 D^n + A_1 D^{n-1} D' + A_2 D^{n-2} D'^2 + \dots + A_n (D')^n$$

replace D by m and D' by 1

$$AE = (A_0 m^n + A_1 m^{n-1} + \dots + A_n) = 0.$$

$$m = m_1, m_2, \dots, m_n$$

$$\text{Then } CF = \phi_1(y + m_1 x) + \phi_2(y + m_2 x) + \phi_3(y + m_3 x) + \dots + \phi_n(y + m_n x)$$

Suppose

$$(D - m_1 D')z = 0.$$

$$\frac{\partial z}{\partial x} - m_1 \frac{\partial z}{\partial y} = 0.$$

$$P - m_1 Q = 0 \quad (P = \frac{\partial}{\partial x}, Q = \frac{\partial}{\partial y})$$

$$\frac{dx}{1} = \frac{dy}{-m_1} = \frac{dz}{0}$$

$$-m_1 dx = dy$$

$$dy + m_1 dx = 0$$

$$(y + m_1 x) = a$$

$$dz = 0$$

$$z = b = f(a) = f(y + m_1 x)$$

$$\text{Let } z = f(y + m_1 x)$$

$$\phi(D, D')z = 0$$

$$\text{Now } Dz = m_1 F'(y + m_1 x)$$

$$D^2 z = m_1^2 F''(y + m_1 x)$$

$$D^n z = m_1^n F^{(n)}(y + m_1 x)$$

$$D'z = F'(y + m_1 x)$$

$$D'^2 z = F''(y + m_1 x)$$

$$(D')^n z = F^{(n)}(y + m_1 x)$$

$$D^r (D')^{n-r} z = \frac{\partial^n z}{\partial x^r \partial y^{n-r}} = m_1^r F^{(n)}(y + m_1 x)$$

$$(A_0 m^n + A_1 m^{n-1} + \dots + A_n) F^{(n)}(y + m_1 x) = 0$$

Q) Solve $ax + ss + at = 0$

$$(aD^2 + sD + a)z = 0.$$

$$\text{putting } D = m, D' = 1 \text{ then } (am^2 + sm + a)z = 0$$

$$AE \quad 2m^2 + sm + 2 = 0$$

$$(m + 1/2)(m + 2) = 0.$$

$$z = \phi(y - 1/2 x) + \psi(y - 2x)$$

$$z = \phi(2y - x) + \psi(2y - 2x) \quad \text{Hence this is the solution}$$

$$\frac{-5 \pm \sqrt{25 - 16}}{4}$$

$$\frac{-5 + 3}{4} \quad \frac{-5 - 3}{4}$$

$$-1/2, -2$$

When the Auxiliary equation has equal roots

Suppose $(D-m)^2 z = 0$ this eqⁿ has 2 equal roots.

$$(D-m)(D-m)z = 0.$$

putting $(D-m)z = u$ — (1)

$$(D-m)u = 0 \quad \text{A.E.}$$

$$p - mq = 0 \quad \checkmark \quad \frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{0}$$

$$y + mx = c_1$$

$$u = c_2 = f(y).$$

$$u = f(y + mx) \quad \text{--- (2)}$$

from (1) & (2)

Then $(D-m)z = f(y + mx)$

$$\frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{f(y + mx)}$$

from 1st two $dy + m dx = 0$
 $y + mx = a$

from 1st and 3rd $\frac{dx}{1} = \frac{dz}{f(y + mx)} = \frac{dz}{f(a)}$

$$z = x f(a) + b = (y + mx) f(a) + f(a)$$

$$z = x f(y + mx) + g(y + mx)$$

Thus in general if the auxiliary equation has r roots equal then

$$z = x^{r-1} f_1(y + mx) + x^{r-2} f_2(y + mx) + \dots + f_r(y + mx)$$

Q) solve $x^2 + 2x + 1 = 0$

Sol: $(x^2 + 2x + 1) = 0$

Q) solve $(D^3 + 6D^2 + 11D + 6)z = 0$

Sol: A.E. : $(m^3 + 6m^2 + 11m + 6) = 0 \quad m = -1, -2, -3$

then solⁿ $z = f_1(y + x) + f_2(y + 2x) + f_3(y + 3x)$

Q) solve $(D^4 - 2D^3 + 2D^2 - D)z = 0$

A.E. $(m^4 - 2m^3 + 2m^2 - m) = 0 \Rightarrow m(m-1)(m-1)^2 = 0$
 $m = 0, 1, 1, 1$

then solⁿ

$$z = f_1(y) + f_2(y + x) + x f_3(y + x) + x^2 f_4(y + x)$$

P.I let the diff eqⁿ $f(D, D')z = \phi(x, y)$ then $P.I = \frac{\phi(x, y)}{f(D, D')}$

Notes: If $\phi(x, y) = x^m y^n$

if $n < m$, $\frac{1}{f(D, D')}$ should be expanded in powers of D'/D

where (ii) if $m < n$ then $\frac{1}{f(D, D')}$ should be expanded in powers of D/D'

Q) Solve $(D^2 - 6DD' + 9D'^2)z = 12x^2 + 36xy$

Sol:

For C.F $D^2 - 6DD' + 9D'^2$

AE $m^2 - 6m + 9$ $m = 3, 3$ then C.F = $f_1(y+3x) + x f_2(y+3x)$

$$P.I = \frac{1}{D^2 - 6DD' + 9D'^2} (12x^2 + 36xy) = \frac{1}{D^2 \left(1 - \frac{6D'}{D} + \frac{9D'^2}{D^2}\right)} (12x^2 + 36xy)$$

$$= \frac{1}{D^2} \left(1 - \left(\frac{6D'}{D} - \frac{9D'^2}{D^2}\right)\right)^{-1} (12x^2 + 36xy)$$

$$= \frac{1}{D^2} \left(1 + \frac{6D'}{D} - \frac{9D'^2}{D^2} + \left(\frac{6D'}{D} - \frac{9D'^2}{D^2}\right)^2 + \dots\right) (12x^2 + 36xy)$$

$$= \frac{1}{D^2} \left(1 + \frac{6D'}{D} - \frac{9D'^2}{D^2} + \frac{36D'^2}{D^2}\right) (12x^2 + 36xy)$$

$$= \frac{1}{D^2} \left(12x^2 + 36xy + \frac{72x(0+36x)}{D} + 27 \frac{D'^2}{D^2} (12x^2 + 36xy)\right)$$

$$= \frac{12x^4}{3 \times 4} + \frac{36y^3}{2 \times 2} + \frac{72 \times 36}{2 \times 3 \times 4} x^4 + \left(\frac{27 \times 12(0+36x)}{D^4}\right)$$

$$= 10x^4 + 6x^3y$$

short methods

When $\phi(x, y) = \phi(ax+by)$

$$D\phi(ax+by) = a\phi'(ax+by)$$

$$D^2\phi(ax+by) = a^2\phi''(ax+by)$$

$$b'D\phi(ax+by) = b\phi'(ax+by)$$

$$D'^2\phi(ax+by) = b'^2\phi''(ax+by)$$

$$f(D, D')\phi(ax+by) = f(a, b)\phi''(ax+by)$$

$$\boxed{\frac{1}{f(D,D')} \varphi^n(ax+by) = \frac{1}{f(a,b)} \varphi(ax+by)}$$

$$f(a,b) \neq 0.$$

$$9) (D^2 - 2DD' + D'^2)Z = e^{x+2y} + x^3.$$

$$AE: m^2 - 2m + 1 \quad m = 1, 1$$

$$\boxed{C.F = \varphi_1(y+x) + x\varphi_2(y+x)}$$

$$P.I = \frac{1}{D^2 - 2DD' + D'^2} e^{x+2y} + \frac{1}{D^2 \left(1 - \frac{2D'}{D} + \frac{D'^2}{D^2}\right)} x^3.$$

$$D=1, D'=2$$

$$= \frac{e^{x+2y}}{D^2} + \frac{1}{D^2} \left(1 - \left(\frac{2D'}{D} - \frac{D'^2}{D^2}\right)\right)^{-1} (x^3)$$

$$= e^{x+2y} + \frac{1}{D^2} \left(1 + \frac{2D'}{D} + \frac{D'^2}{D^2}\right) \left(\frac{4D'^2}{D^2} - \frac{4D'^3}{D^3}\right) x^3.$$

$$= e^{x+2y} + \frac{1}{D^2} (x^3) = e^{x+2y} + \frac{x^5}{4 \cdot 15}$$

exceptional cases when $f(a,b) = 0$.

$$\text{suppose } (bD - aD')Z = x^r \varphi(ax+by)$$

$$\text{Then } bD - aD' = x^r \varphi(ax+by)$$

$$\frac{dx}{b} = \frac{dy}{-a} = \frac{dz}{x^r \varphi(ax+by)}.$$

$$\text{Taking 1st two } ax+by = c$$

$$\text{from 1st \& 3rd}$$

$$dz = \frac{x^r \varphi(c) dx}{b} \Rightarrow z = \frac{x^{r+1}}{b(r+1)} \varphi(c) = \frac{x^{r+1}}{b(r+1)} \varphi(ax+by) \quad **$$

$$Z = \frac{1}{(bD - aD')^n} x^r \varphi(ax+by) = \frac{x^{r+1}}{b(r+1)} \varphi(ax+by).$$

$$\text{Hence } Z = \frac{1}{(bD - aD')^n} \varphi(ax+by)$$

$$= \frac{1}{(bD - aD')^{n-1}} \cdot \frac{1}{(bD - aD')} x^r \varphi(ax+by)$$

$$= \frac{1}{(bD - aD')^{n-2}} \frac{x^2}{2b^2} \varphi(ax+by)$$

$$\frac{1}{(bD-ad)^n} \psi(ax+by) = \frac{x^n}{n! b^n} \psi(ax+by)$$

Q) Solve $(4D^2 - 4DD + D^2)z = 16 \log(x+2y)$
 $\downarrow$
 $(2D-D)^2 z = 16 \log(x+2y)$

AE $2m-1=0$ C.F = $f_1(2y+x) + x f_2(2y+x)$

P.I $\frac{1}{(2D-D)^2} 16 \log(x+2y) = \frac{16x^2}{2!(2)^2} \log(x+2y)$

$\boxed{Z = C.F + P.I}$ $z = f_1(2y+x) + x f_2(2y+x) + \frac{x^2}{2} \log(x+2y)$

Since

$$\frac{1}{f(D,D')} \psi^n(ax+by) = \frac{1}{f(a,b)} \psi^n(ax+by)$$

Suppose $ax+by=v$ $\frac{1}{f(D,D')} \psi^n(v) = \frac{1}{f(a,b)} \psi^n(v)$

$$\frac{1}{f(D,D')} \psi(v) = \frac{1}{f(a,b)} \int \int \int \dots \int \psi(v) dv \dots dv$$

n terms.

General method:

$$(D-mD')z = \phi(x,y)$$

$$P-mQ = \phi(x,y) \Rightarrow \frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{\phi(x,y)}$$

$\downarrow$
y+mx=a

from 1st & 3rd

$$dz = \phi(x,y) dx$$

$$z = \int \phi(x, a-mx) dx$$

$$P.I = \frac{1}{D-mD'} \phi(x,y) = \int \phi(x, a-mx) dx$$

Q) solve $x+6-6t = y \cos x$

C.F = - - - -

$$P.I = \frac{1}{(D+3D')(D-2D)} y \cos x = \frac{1}{(D+3D')} \int \cos x \cdot (a-2x) dx = \frac{1}{D+3D'} (y \sin x + 2x \cos x)$$

$$= \frac{1}{(D+3D')} (y \sin x + 2x \cos x)$$

Q) solve $(2D^2 - 5DD' + 2D'^2)z = 24(y-x)$

Sol: for C.F $m^2 - 5m + 2 = 0 \quad \therefore m = \frac{5 \pm \sqrt{25-16}}{4} = \frac{5+3}{4}, \frac{5-3}{4}$
 $= 2, 1/2$

C.F = $f_1(y+2x) + f_2(2y+x)$

P.I = $\frac{1}{(2D^2 - 5DD' + 2D'^2)} 24(y-x)$

$= \frac{1}{2(-1)^2 - 5(-1)(1) + 2(1)^2} \iint v dv dv, v = y-x$

$= \frac{24}{9} \int \frac{v^2}{2} dv = \frac{24}{9} \cdot \frac{v^3}{3} = \frac{4}{9} (y-x)^3$

$\frac{1}{f(D,D')} \phi(v) = \frac{1}{f(a,b)} \phi(v)$
 $\frac{1}{f(D,D')} \phi(v) = \frac{1}{f(a,b)} \int \dots \int dv dv$
 $v = ax+by$

Non-homogeneous linear PDE with const coeff:

$(D - mD' - \alpha)z = 0$

$p - mq = \alpha z$

$\frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{\alpha z}$

$y+mx = a(\text{const})$

from 1st & 3rd

$\frac{dz}{z} = \alpha dx$

$\log z = \alpha x + \log b$

$z = e^{\alpha x} \cdot b = e^{\alpha x} \cdot f(a)$

$\boxed{z = e^{\alpha x} \cdot f(y+mx)}$

Let $\phi(D,D')z = (D - m_1D' - \alpha_1)(D - m_2D' - \alpha_2) \dots (D - m_nD' - \alpha_n)z = 0$

then

$z = e^{\alpha_1 x} f_1(y+m_1x) + e^{\alpha_2 x} f_2(y+m_2x) + \dots + e^{\alpha_n x} f_n(y+m_nx)$

If $\phi(D,D')z = (D - mD' - \alpha)z = 0$

$z = e^{\alpha_1 x} f_1(y+m_1x) + x e^{\alpha_2 x} f_2(y+m_2x) + x^2 e^{\alpha_3 x} f_3(y+m_3x) + \dots + x^{r-1} e^{\alpha_r x} f_r(y+m_rx)$

P.I

Q) $\frac{1}{f(D,D')} e^{ax+by} = \frac{1}{f(a,b)} e^{ax+by}, f(a,b) \neq 0$

i.e put $D=a, D'=b$

(ii) If RHS of diff eqn $\sin(ax+by)$ or $\cos(ax+by)$ then we separate

$$\frac{1}{-P(D^2, D, D^2)} \sin(ax+by) \text{ or } \cos(ax+by) = \frac{1}{f(-a^2, (-a)b, (+b)^2)} \sin(ax+by) \text{ or } \cos(ax+by)$$

(iii) $\frac{1}{f(D, D)} x^m y^n = [f(D, D)]^{-1} x^m y^n$

iv) $\frac{1}{f(D, D)} e^{ax+by} = e^{ax+by} \frac{1}{f(D+a, D+b)} V$ V is any function of xy

Q) Solve $(D-3D'-2)^2 z = 2e^{2x} \tan(y+3x)$

G.F = $e^{2x} f_1(y+3x) + x e^{2x} f_2(y+3x)$

P.I = $\frac{1}{(D-3D'-2)^2} 2e^{2x} \tan(y+3x)$

P.I = $2e^{2x} \frac{1}{(D+2-3D'-2)^2} \tan(y+3x)$

P.I = $2e^{2x} \frac{1}{(D-3D')^2} \tan(y+3x) = 2e^{2x} \frac{x^2}{2 \times 1} \tan(y+3x)$

Method of Separation of Variables

In this method we assume that the dependent variable is the product of function of independent variable. So the diffⁿ eqn are solved as equal to no. of independent variables.

Q) Using method of separation of variables solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$, where
 $u(x, 0) = 6e^{-3x}$

Sol: Let $u = X(x)T(t) \rightarrow (1)$

$\frac{\partial}{\partial x} X T = 2 \frac{\partial}{\partial t} X T + X T$

$$T \frac{dx}{dt} = 2x \frac{dT}{dt} + xT$$

$$(1) \quad T x' = 2x T' + xT$$

$$(2) \quad \frac{x'}{x} = 2 \frac{T'}{T} + 1 = c \text{ (say)} \rightarrow (3)$$

$$a) \quad \frac{x'}{x} = c \quad (1) \quad \frac{1}{x} \frac{dx}{dt} = c$$

$$(2) \quad \frac{dx}{x} = c dt$$

$$\log x = cx + \log a$$

$$\underline{x = a e^{cx}}$$

$$b) \quad 2 \frac{T'}{T} + 1 = c$$

$$2 \frac{dT}{T} + 1 = c$$

$$\frac{dT}{T} = \frac{1}{2}(c-1) dt$$

$$\log T = \frac{1}{2}(c-1)t + \log b$$

$$\log \frac{T}{b} = \frac{1}{2}(c-1)t$$

$$\boxed{T = b e^{\frac{1}{2}(c-1)t}}$$

$$\text{from (2)} \quad u(x,t) = a b e^{cx + \frac{1}{2}(c-1)t}$$

$$\text{If } t=0, u(x,0) = 6 e^{-3x}, \text{ then}$$

$$6 e^{-3x} = a b e^{cx} \Rightarrow ab = 6, \quad c = -3$$

$$\boxed{u(x,t) = 6 e^{-3x-2t}}$$

Q) Solve $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$ Solve this by method of separable variables

sol: $z = X(x)Y(y)$

$$\text{then } Y \frac{d^2 X}{dx^2} - 2Y \frac{dX}{dx} + X \frac{dY}{dy} = 0$$

$$Y(X'' - 2X') = -XY'$$

$$\Rightarrow \frac{X'' - 2X'}{X} = -Y'/Y = a \text{ (say)}$$

$$X'' - 2X' - aX \quad \& \quad Y' + aY = 0$$

Solve $\downarrow$
 $\frac{d^2 X}{dx^2} - 2 \frac{dX}{dx} - aX = 0$

$$\text{AE: } D^2 - 2D - a = 0$$

$$D = 1 \pm \sqrt{1+a}$$

$$X = c_1 e^{(1+\sqrt{1+a})x} + c_2 e^{(1-\sqrt{1+a})x}$$

Also $y' + ay = 0$

$$\frac{dy}{y} + a dy = 0$$

$$\log y + ay = \log c$$

$$\log y/c = -ay$$

$$y = ce^{-ay}$$

$$z = xy$$

A eqⁿ reducible to linear eqⁿ with const coefficients

let the PDE of the form $a_0 x^n \frac{\partial^n z}{\partial x^n} + a_1 x^{n-1} y \frac{\partial^n z}{\partial x^{n-1} \partial y} + a_2 x^{n-2} y^2 \frac{\partial^n z}{\partial x^{n-2} \partial y^2} + \dots$

$$a_n y^n \frac{\partial^n z}{\partial y^n} = f(x, y)$$

Suppose $x = e^x, y = e^y$

or $\log x = x, \log y = y$

Then $x \frac{\partial}{\partial x} = D$

$$x^2 \frac{\partial^2}{\partial x^2} = D(D-1)$$

$$x^n \frac{\partial^n}{\partial x^n} = D(D-1)(D-2) \dots (D-n+1)$$

$$y \frac{\partial}{\partial y} = D_2$$

$$y^2 \frac{\partial^2}{\partial y^2} = D_2(D_2-1)$$

$$x^m y^n D^m D_2^n = D(D-1) \dots (D-m+1) D_2(D_2-1) \dots (D_2-n+1)$$

Q) Solve $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = 0$

Solⁿ: put $e^x = x, e^y = y \quad \frac{\partial}{\partial x} = D, \frac{\partial}{\partial y} = D_2$

$$[D(D-1) + 2DD_2 + D_2(D_2-1)]z = 0$$

$$(D+D_2)(D+D_2-1)z = 0$$

$$C.F = e^{ax} f(y-x) + e^{bx} g(y-x)$$

$$= f(\log y - \log x) + x g(\log y - \log x)$$

$$= f_1(y/x) + x g_1(y/x)$$